Annex B

Computational Thinking Exercise: "Smart Vending Machine"

Section: Lithium **Score:____________**
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Scenario

Your school installs a vending machine to provide snacks and drinks. However, students encounter several issues:

- Sometimes the machine does not give the correct change.
- Items run out, but the machine doesn't notify anyone.
- Students press the wrong buttons and get the wrong item.
- The machine is slow when multiple students use it in succession.

Your task is to **decompose this problem** into smaller, manageable parts that could be solved with computational thinking (CT) Skills.

Step 1: Identify the Big Problem

Main Problem: The vending machine has problems within its codes. Specifically in its system efficiency, change calculation, input reception, and item counting.

Step 2: Identify three to four Sub-Problems

Please list possible sub-problems:

1. The machine systems do not keep track of how many of each item are left. It also does not include a notification system that informs the user that the item is unavailable.

2. The selection buttons of the vending machine correspond to the wrong item and possibly display the wrong number.

3. The machine incorrectly calculates and gives change, which may be due to an error in the code for calculating the amount of change.

4. The machine's code is likely inefficient or contains many errors, which leads to its performance slowing down or stalling.

Step 3: Define Computational Thinking Approaches

For each sub-problem, apply CT skills:

Sub-Problem	CT Skill	Example Solution

Step 4: Draw a flowchart or write a pseudocode for the identified sub-problem (Your group could use a separate sheet of paper)

Rubrics For Grading

Total Points: 20pts

Criteria & Levels of Performance

Criteria	Excellent (4)	Good (3)	Fair (2)	N.I. (1)
Identification of Sub-Problems	Identifies 3+ clear, relevant sub-problems that directly connect to the scenario.	Identifies 2–3 mostly relevant sub-problems.	Identifies 1–2 vague or partially relevant sub-problems.	Struggles to identify sub-problems or lists unrelated issues.
Application of CT Strategies	Correctly applies appropriate CT strategies (abstraction, decomposition, pattern recognition, algorithm design) to each sub-problem with clear reasoning.	Applies CT strategies to most sub-problems, with minor errors or limited explanation.	Applies CT strategies inconsistently, with weak or unclear reasoning.	Rarely applies CT strategies or misuses them.
Flowchart / Pseudocode X 2	Flowchart / Pseudocode is complete, logical, and easy to follow; shows clear steps and decision points.	Flowchart / Pseudocode is mostly complete and logical, with minor gaps or unclear steps.	Flowchart / Pseudo Code is partially complete, missing key steps or connections.	Flowchart / Pseudocode is incomplete, confusing, or missing entirely.
Reflection / Explanation	Provides thoughtful reflection on how decomposition helps problem-solving and identifies CT skills used with strong justification.	Provides adequate reflection with some justification of CT skills.	Provides limited reflection with weak or generic justification.	Provides minimal or no reflection.